Propose an interfacial treatment and treatment method for the homogeneous dispersion of nano-TiO2 in NBR rubber and identify a suitable interfacial treatment method

Interfacial Treatment Methods for Homogeneous Dispersion of Nano-TiO_2 in NBR Rubber

1. Introduction

The homogeneous dispersion of nano-TiO_2 in nitrile butadiene rubber (NBR) is a critical challenge that determines the ultimate performance of the composite. Nano-TiO_2 particles have a strong tendency to agglomerate due to their high surface energy and polar surface chemistry, while NBR is a non-polar to moderately polar elastomer. This incompatibility leads to poor dispersion, weak filler-rubber interaction, and compromised mechanical and dielectric properties . Effective interfacial treatment is therefore essential to overcome this challenge and fully realize the reinforcing potential of nano-TiO_2 in NBR composites.

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2. Proposed Interfacial Treatment Methods

Based on the available literature, several interfacial treatment approaches have been developed to improve nano-TiO_2 dispersion in NBR. These methods can be categorized into surface modification strategies and in-situ synthesis approaches.

2.1 Surface Modification with Silane Coupling Agents

Silane coupling agents are bifunctional molecules that can chemically bond to the TiO_2 surface through their hydrolyzable groups while providing organic functional groups that enhance compatibility with the NBR matrix.

KH550 (γ-aminopropyltriethoxysilane): This silane coupling agent has been successfully used to modify TiO_2 nanoparticles for NBR composites . The modification improves surface hydrophobicity and reduces particle agglomeration through chemical bonding.

KH570 (γ-methacryloxypropyltrimethoxysilane): Combined with polydopamine (PDA) coating, KH570 grafting creates a functionalized TiO_2 surface that significantly improves filler dispersion in NBR, leading to higher dielectric constant, lower elastic modulus, and enhanced electric breakdown field .

TESPT (bis-(3-triethoxysilylpropyl)-tetrasulfide): When added during in-situ sol-gel titania generation, TESPT causes further improvement in filler dispersion and composite properties compared to external addition during mixing .

2.2 Surface Modification with Titanate Coupling Agents

NDZ201 (titanate coupling agent): This coupling agent has been demonstrated to be superior to KH550 for TiO_2 modification, as it creates a more effective organic molecular layer on the particle surface, preventing aggregation and improving compatibility with XNBR (carboxylated NBR) . NDZ201-modified TiO_2 exhibited a contact angle of 90.4°, indicating significant hydrophobicity enhancement.

2.3 Bio-Inspired Polydopamine (PDA) Coating

Inspired by mussel adhesive proteins, dopamine self-polymerization is used to deposit a thin adherent PDA film onto TiO_2 surfaces. The PDA layer:

- Improves interfacial interaction between TiO_2 and NBR

- Provides chemisorption sites for further functionalization

- Results in better filler dispersion, reduced dielectric loss, and higher breakdown strength

2.4 Combined Covalent and Noncovalent Modification

PCPA-GPTMS Modification: TiO_2 nanoparticles are first modified by noncovalent deposition of poly(catechol/polyamine) (PCPA) and subsequently functionalized with γ-(2,3-epoxypropoxy)-propytrimethoxysilane (GPTMS) . This combined approach:

- Preserves the original surface characteristics of TiO_2 nanoparticles

- Introduces epoxy groups that provide strong interfacial interaction with polar NBR

- Enhances mechanical, dielectric, and electromechanical properties

- Is cheaper and requires shorter reaction times compared to dopamine modification

2.5 Sodium Polyacrylate (PAAS) Modification

PAAS, a water-soluble polymer with negative charge, modifies TiO_2 through charge mutual repulsion mechanisms. PAAS-modified TiO_2 shows good dispersibility with low agglomeration due to the formation of a molecular layer on the particle surface .

2.6 In-Situ Sol-Gel Synthesis

Instead of externally mixing pre-formed TiO_2, nano-titania can be synthesized directly in the NBR solution using titanium n-butoxide (TNB) as a precursor . This in-situ approach:

- Achieves very good dispersion of nanometric titania

- Enhances rubber-filler interaction

- Results in significantly higher mechanical properties than externally filled composites

- Enables controlled incorporation of titania under optimized reaction conditions

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3. Identification of the Most Suitable Interfacial Treatment Method

Based on the comprehensive analysis of available methods, the **Combined Covalent and Noncovalent Modification (PCPA-GPTMS)** approach emerges as the most suitable interfacial treatment method for the following reasons:

3.1 Superior Performance Enhancement

The PCPA-GPTMS modification achieves a remarkable actuated strain of approximately 11% at 47 kV/mm, which is about **230% of pure NBR** (4.8%) . This demonstrates significant enhancement in electromechanical properties compared to other modification approaches.

3.2 Cost-Effectiveness and Efficiency

Compared to dopamine modification, the PCPA-GPTMS method is **cheaper and requires shorter reaction times**, making it more practical for industrial-scale applications .

3.3 Balanced Modification Strategy

The method combines noncovalent deposition (preserving surface characteristics) with covalent functionalization (providing strong interfacial interaction), creating a comprehensive solution to the dispersion challenge.

3.4 Versatility

This approach has been validated for enhancing mechanical, dielectric, and electromechanical properties of NBR composites, making it suitable for diverse applications .

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4. Treatment Procedure for the PCPA-GPTMS Modification Method

Step 1: PCPA Noncovalent Deposition

- Prepare TiO_2 nanoparticles
- Deposit poly(catechol/polyamine) (PCPA) onto TiO_2 surfaces through noncovalent interactions
- This step preserves the original surface characteristics of TiO_2 nanoparticles

Step 2: GPTMS Covalent Functionalization

- Functionalize PCPA-coated TiO_2 with γ-(2,3-epoxypropoxy)-propytrimethoxysilane (GPTMS)
- The epoxy groups of GPTMS react with hydroxyl and amine groups on the PCPA surface
- This creates strong interfacial interaction sites with the polar NBR matrix

Step 3: Composite Preparation

- Incorporate modified TiO_2-PCPA-GPTMS nanoparticles into NBR
- The optimal loading is typically around 10 phr for balanced property enhancement
- Mix using standard rubber compounding techniques

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5. Mechanism of the Recommended Treatment

The effectiveness of the PCPA-GPTMS modification relies on a dual-interaction mechanism:

5.1 Noncovalent Component (PCPA)

- Forms a uniform coating on TiO_2 surfaces without altering intrinsic particle properties
- Maintains particle stability and prevents agglomeration
- Provides reactive sites (amine groups) for subsequent covalent bonding

5.2 Covalent Component (GPTMS)

- Epoxy groups form strong chemical bonds with the PCPA layer
- Creates a robust organic shell that is chemically compatible with NBR
- The epoxy groups can react with polar -C≡N groups in NBR, creating strong interfacial adhesion

5.3 Synergistic Effect

The combination of noncovalent and covalent modification:

- Maximizes filler dispersion through steric stabilization
- Enhances interfacial adhesion through chemical bonding
- Improves stress transfer from matrix to filler
- Results in superior mechanical and electromechanical properties

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6. Alternative Recommended Method: In-Situ Sol-Gel Synthesis

If surface modification is not preferred, the **in-situ sol-gel synthesis** method offers a promising alternative :

Procedure:

1. Dissolve NBR in an appropriate solvent

2. Add titanium n-butoxide (TNB) as the titania precursor

3. Control hydrolysis and condensation reactions under optimized conditions

4. Generate nano-titania directly within the NBR solution

5. Optionally add TESPT silane coupling agent during in-situ generation for further property enhancement

Advantages:

- Superior dispersion compared to external mixing

- Enhanced rubber-filler interaction

- Significantly higher mechanical properties

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7. Summary

| Treatment Method | Key Mechanism | Best For | Performance Highlights |

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| **PCPA-GPTMS** | Noncovalent + covalent | Electromechanical properties | 230% increase in actuated strain |

| PDA-KH570 | Bio-inspired + silane | Dielectric properties | 53% larger actuated strain |

| TESPT in-situ | Sol-gel + coupling | Mechanical reinforcement | Superior to external mixing |

| KH550/NDZ201 | Silane/titanate coupling | General dispersion | Good hydrophobicity |

| PAAS | Charge repulsion | Colloidal dispersion | Low agglomeration |

Recommended Method: The **PCPA-GPTMS combined covalent and noncovalent modification** is identified as the most suitable interfacial treatment method due to its superior performance, cost-effectiveness, and practical advantages.

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